50.051: Programming Languages Concept - Introduction

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```
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <arpa/inet.h>

void serveur1(portServ ports)
{
    int sockServ1, sockServ2, sockClient;
    struct sockaddr_in monAddr, addrClient, addrServ2;
    socklen_t lenAddrClient;

    if ((sockServ1 = socket(AF_INET, SOCK_STREAM, 0)) == -1) {
    perror("Erreur socket");
    exit(1);
    }
    if ((sockServ2 = socket(AF_INET, SOCK_STREAM, 0)) == -1) {
    perror("Erreur socket");
    exit(1);
    }

    bzero(&monAddr, sizeof(monAddr));
    monAddr.sin_family = AF_INET;
    monAddr.sin_port = htons(ports.port1);
    monAddr.sin_addr.s_addr = INADDR_ANY;
    bzero(&addrServ2, sizeof(addrServ2));
```

Instructor: Simon Perrault

- Lecturer: Simon PERRAULT
  - Assistant Professor
  - Research area: Human Computer Interaction / Human AI Collaboration
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  - Teaching: weeks 1, 2, 3, 4, 6, 8

Instructor: Matthieu De Mari

- Lecturer: Matthieu DE MARI
  - Senior Lecturer
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  - Teaching: weeks 5, 9-13

Course Handout

- Everything you need to know about the course
- https://docs.google.com/document/d/15hx3jF3vzdnOw7yyjJGTyliBZfFbvOseiFiou8xwRRg/edit?usp=sharing
- **Before you ask any question, check the handout.**

What is it about?

- Part 1: C programming + State Machines (6 weeks)
  - Everything about C
  - Memory Management
  - State Machines
- Part 2: Compilation & C++ (6 weeks)
  - Regular Expressions
  - Lexical/Syntaxic/Semantic Analysis
  - Parsing
  - C++

PART 1: C

About C

- C created in 1972 by Dennis Ritchie between 69-73
- Called C as a successor to B
- Imperative programming language
  - Program = set of statements
- C inspired:
  - C++
  - Objective-C
  - Java
  - C#
  - PHP

Restricted

Why should I learn C?

- C used for operating systems
  - Unix
  - Linux
  - Windows
  - Mac OS X
- Can be compiled on nearly any hardware
- “Low level language”
  - Memory management
  - Small set of keywords

Hilarious Quotes about C

- “`Students can learn C in 2 weeks in a normal module`” – Anonymous (former EPD HoP)
- “`I am good in C,… except pointers`” – too many people

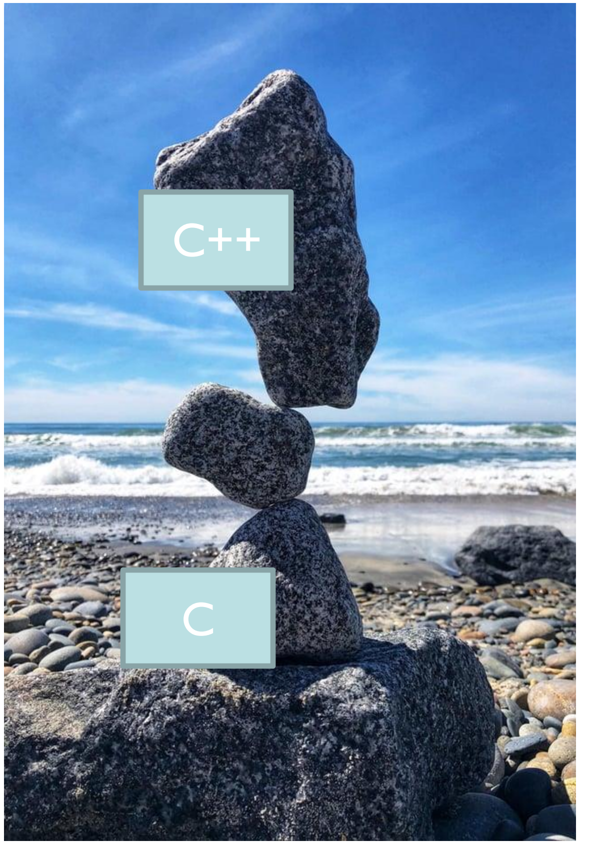
PART 2: COMPILATION

Compilation

- What is compilation?
- How to go from Code to Assembly
  - Tokenization
  - Lexical Analysis
  - Syntaxic Analysis
  - Semantic Analysis
  - Parsing

C++

- Created in 1985
- ”C with Classes”
- Kinda like Java
  - More “powerful”?
  - More complicated
- **If you know Java and C++, you know C#**



FINAL GRADE

Final Grade

- Practicals (20%)
  - 6 individual assignments
  - Due 2 weeks after session
- Group Project (20%)
  - 4 students per group max, can create own group
  - Instructions given in week 2
- Class participation + Feedback (8+2%)
- Midterm Exam (20%)
- Final Exam (30%)

Questions?